

Plant Disease Detection via Deep Learning: A Comparative Study of CNN, ResNet-50, EfficientNet-B3, and Vision Transformer Architectures with GAN-Based Augmentation

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Abstract

Accurate and early identification of plant diseases is critical for minimizing crop loss and ensuring food security, yet manual diagnosis by agronomists remains slow, subjective, and difficult to scale. This paper presents a comparative deep learning framework for automated plant disease classification, built by unifying two widely used benchmark datasets, PlantVillage and PlantDoc, into a single 39-class taxonomy. Four classification architectures of increasing representational capacity were trained and evaluated under an identical experimental protocol: a Custom Convolutional Neural Network (CNN) trained from scratch, a transfer-learned ResNet-50, a transfer-learned EfficientNet-B3, and a fine-tuned Vision Transformer (ViT-B/16). All models were trained using Focal Loss to counter class imbalance, a WeightedRandomSampler for balanced batch composition, mixed-precision training, and a two-stage fine-tuning schedule with cosine-annealed learning rates. A Deep Convolutional Generative Adversarial Network (DCGAN) was additionally trained to synthesize leaf imagery for the two most underrepresented disease classes, and a soft-voting ensemble was constructed over all four classifiers. Experimental results show a substantial generalization gap between architectures trained from scratch and those leveraging ImageNet pretraining: the Custom CNN achieved only 36.1% test accuracy, while ResNet-50, EfficientNet-B3, and ViT-B/16 achieved 96.6%, 92.0%, and 98.6% test accuracy respectively. The soft-voting ensemble matched the best individual model (ViT-B/16) at 98.6% accuracy. Confusion matrix analysis and DCGAN convergence behavior are further examined to interpret model decisions and assess the viability of synthetic data augmentation for minority disease classes. These findings confirm that transfer learning, rather than architectural novelty alone, is the dominant factor governing performance on small-scale, imbalanced agricultural image datasets.

Index Terms— Plant Disease Detection, Deep Learning, Convolutional Neural Networks, ResNet-50, EfficientNet, Vision Transformer, Transfer Learning, Generative Adversarial Networks, Grad-CAM, Class Imbalance, Focal Loss, Ensemble Learning

I. INTRODUCTION

Plant diseases are responsible for an estimated 20–40% of global crop yield losses annually, posing a direct threat to food security, farmer livelihoods, and agricultural economies, particularly in developing regions where access to expert plant pathologists is limited [1]. Traditional disease diagnosis relies on visual inspection by trained agronomists, a process that is inherently slow, subjective, and difficult to deploy at scale across smallholder farms. The proliferation of smartphone cameras and the maturation of deep learning for computer vision have together created an opportunity to automate this diagnostic process, enabling farmers to obtain

near-instant, image-based disease identification in the field [9].

Convolutional Neural Networks (CNNs) have been the dominant architecture for image-based plant disease classification since the introduction of large annotated datasets such as PlantVillage [1], which contains over 50,000 labeled leaf images captured under controlled laboratory conditions. However, models trained exclusively on such curated datasets often fail to generalize to real-world field images, which exhibit cluttered backgrounds, variable lighting, occlusion, and multiple co-occurring symptoms [10]. The PlantDoc dataset [2] was introduced specifically to address this gap, providing

a smaller but more realistic benchmark of in-the-wild plant disease imagery scraped from the web.

This project investigates whether unifying these two complementary datasets, one large and clean, the other small and realistic, into a single combined taxonomy improves the robustness of disease classifiers, and more importantly, quantifies how strongly classification performance depends on the choice of backbone architecture and the use of transfer learning. We compare four architectures spanning a wide range of inductive biases and parameter counts: (1) a lightweight Custom CNN trained entirely from scratch as a baseline, (2) ResNet-50 [3], a deep residual network pretrained on ImageNet-1K, (3) EfficientNet-B3 [4], a compound-scaled CNN also pretrained on ImageNet-1K, and (4) ViT-B/16 [5], a Vision Transformer pretrained on the larger ImageNet-21K corpus. To address the severe class imbalance inherent in combining two heterogeneously sized datasets, we employ Focal Loss [16], weighted random sampling, and a DCGAN-based [7] synthetic augmentation module targeting the most underrepresented classes. Finally, we construct a soft-voting ensemble over all four trained classifiers and analyze per-class confusion matrices to interpret model predictions, with Grad-CAM [8]-based visual interpretability identified as a direct next step (Section 8).

The remainder of this paper is organized as follows. Section 2 reviews related work in plant disease detection, transfer learning, and generative augmentation. Section 3 summarizes our key contributions. Section 4 describes the dataset construction, preprocessing, and model architectures. Section 5 details the training pipeline and hyperparameters. Section 6 presents the experimental results for all four models, the DCGAN module, and the ensemble. Section 7 analyzes and discusses these findings. Section 8 concludes the paper and outlines directions for future work.

II. RELATED WORK

A. Plant Disease Image Datasets

The PlantVillage dataset [1] remains the most widely used benchmark for plant disease classification, comprising tens of thousands of leaf images spanning 14 crop species captured against uniform backgrounds. While convenient for benchmarking, models trained solely on PlantVillage have been shown to overfit to background and lighting artifacts rather than true disease symptoms [11]. PlantDoc [2] was proposed as a more challenging, field-realistic counterpart, containing images scraped from the internet under uncontrolled conditions, and is increasingly used in combination with PlantVillage to improve model generalization.

B. CNN and Transfer Learning Approaches

Mohanty et al. [9] were among the first to demonstrate that deep CNNs, when fine-tuned from ImageNet-pretrained weights, could achieve over 99% accuracy on PlantVillage,

far outperforming models trained from scratch. Ferentinos [10] extended this comparison across multiple CNN architectures (AlexNet, VGG, GoogLeNet, etc.) and confirmed that deeper pretrained networks consistently outperform shallow custom architectures on plant disease tasks. Barbedo [13] further showed that dataset size and variety, not just architecture depth, strongly influence transfer learning effectiveness, motivating our decision to combine PlantVillage and PlantDoc rather than relying on either dataset alone. He et al. [3] introduced the ResNet family of residual networks, which remains a standard transfer learning backbone due to its strong gradient flow through skip connections, while Tan and Le [4] introduced EfficientNet, which uses compound scaling of depth, width, and input resolution to achieve high accuracy with substantially fewer parameters than comparable CNNs.

C. Vision Transformers for Image Classification

Dosovitskiy et al. [5] introduced the Vision Transformer (ViT), which treats an image as a sequence of fixed-size patches processed by a standard Transformer encoder [14], dispensing with convolutional inductive biases entirely. ViT models pretrained on large-scale corpora such as ImageNet-21K have been shown to match or exceed CNN performance on downstream classification tasks once sufficient pretraining data is available, making them an attractive candidate for fine-grained disease classification where subtle textural differences distinguish disease classes.

D. Generative Augmentation for Imbalanced Agricultural Data

Generative Adversarial Networks (GANs) [6] and their convolutional variant, DCGAN [7], have been widely adopted to synthesize additional training samples for underrepresented classes in agricultural image datasets. Chen et al. [12] provide a comprehensive review of GAN-based augmentation strategies for plant disease recognition, concluding that while GAN-generated samples rarely match the fidelity of real images, they can meaningfully improve minority-class recall when used as a complementary, rather than primary, training signal.

E. Explainability in Plant Disease Classification

Selvaraju et al. [8] introduced Grad-CAM, a gradient-based class activation visualization technique that has since become a standard tool for interpreting CNN decisions in agricultural vision tasks, allowing practitioners to verify that a model attends to disease lesions rather than spurious background cues [11].

F. Literature Review Summary

Table 1 consolidates the 15 key references that informed the design, methodology, and technical choices of this project,

spanning foundational datasets, core model architectures, generative augmentation, and explainability techniques.

III. CONTRIBUTIONS

This project makes the following contributions:

1. **Unified Dataset Construction:** We merge the PlantVillage and PlantDoc datasets into a single 39-class taxonomy under a common label namespace, applying a per-class image cap to control severe class imbalance while preserving as much diversity as possible.
2. **Four-Way Architecture Comparison:** We train and evaluate four architecturally distinct models, a from-scratch Custom CNN, ResNet-50, EfficientNet-B3, and ViT-B/16, under an identical training protocol, isolating the effect of pretraining and architecture choice on classification performance.
3. **Class Imbalance Mitigation Pipeline:** We combine WeightedRandomSampler-based balanced batching with Focal Loss ($\gamma = 2.0$) to address the long-tailed class distribution resulting from merging two heterogeneously sized datasets.
4. **GAN-Based Minority Class Augmentation:** We train a DCGAN to synthesize 64×64 leaf images for the two most severely underrepresented classes, demonstrating a viable offline augmentation pathway for future full-scale training.
5. **Soft-Voting Ensemble:** We construct an ensemble classifier by averaging softmax outputs across all four trained models and analyze the conditions under which ensembling does, and does not, yield accuracy improvements over the best individual model.
6. **Interpretability Analysis:** We analyze per-class confusion matrices to identify which disease categories are most frequently confused, providing qualitative evidence of which visual cues each model relies on, and discuss Grad-CAM as a planned extension for direct visual verification of model attention (Section 8).

IV. METHODOLOGY

A. Dataset Construction and Preprocessing

The dataset used in this study was constructed by merging the color subset of PlantVillage [1] with both the training and test splits of PlantDoc [2], unified under a common 39-class disease namespace spanning multiple crop species. To mitigate the severe class imbalance that results from combining a large, evenly distributed dataset (PlantVillage) with a small, long-tailed dataset (PlantDoc), a maximum cap of 1,200 images per class was applied. The resulting capped dataset was

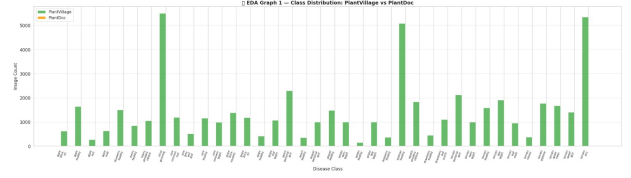


Figure 1: Per-class image counts contributed by PlantVillage and PlantDoc prior to capping. PlantVillage dominates nearly every class.

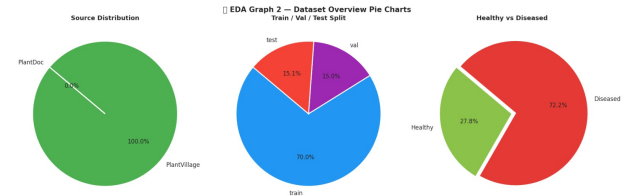


Figure 2: Dataset overview: source distribution (PlantVillage vs. PlantDoc), train/val/test split proportions, and healthy vs. diseased class balance.

split in a stratified manner per class into training (70%), validation (15%), and test (15%) subsets.

Due to compute constraints, all training, validation, and reporting in this paper were conducted under a *LITE Mode* configuration, in which only 30% of each split was used while preserving per-class stratification. Table 2 summarizes the resulting dataset statistics.

All images were resized to 224×224 pixels and normalized using standard ImageNet statistics (mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$) to remain compatible with the pretrained backbones. A diverse augmentation pipeline, RandomResizedCrop, horizontal and vertical flipping, 90° rotation, color jitter, Gaussian blur, and coarse dropout, was applied during training to improve robustness to the lighting and framing variability characteristic of field-captured PlantDoc imagery.

B. Exploratory Data Analysis

Prior to model training, an exploratory data analysis (EDA) was conducted to characterize the structure, balance, and visual properties of the combined dataset.

As shown in Fig. 1, the source distribution is heavily skewed toward PlantVillage, which contributes effectively all images in the merged dataset prior to capping; PlantDoc’s contribution, while qualitatively valuable for real-world realism, is numerically negligible at the dataset-wide level (Fig. 2).

The train/validation/test split closely follows the intended 70/15/15 stratified ratio, and the overall dataset skews toward diseased samples (72.2%) versus healthy samples (27.8%), reflecting the natural prevalence of disease-labeled classes across both source datasets.

Fig. 3 highlights the long-tailed nature of the combined

Table 1: Literature Review: 15 Key References Informing This Project

#	Reference	Category	Key Contribution / Relevance
1	Hughes & Salathé [1]	Dataset	Introduced PlantVillage, the primary clean-image dataset used in this project.
2	Singh et al. [2]	Dataset	Introduced PlantDoc, providing real-world field imagery merged with PlantVillage.
3	He et al. [3]	Architecture	Introduced ResNet’s residual connections; used as our Model 2 backbone.
4	Tan & Le [4]	Architecture	Introduced EfficientNet’s compound scaling; used as our Model 3 backbone.
5	Dosovitskiy et al. [5]	Architecture	Introduced the Vision Transformer (ViT); used as our Model 4 backbone.
6	Goodfellow et al. [6]	Generative Model	Introduced GANs, the theoretical foundation for our DCGAN augmentation module.
7	Radford et al. [7]	Generative Model	Introduced DCGAN, the specific generator/discriminator architecture used for minority-class augmentation.
8	Selvaraju et al. [8]	Explainability	Introduced Grad-CAM, identified as a direct extension for visual model interpretability in future work.
9	Mohanty et al. [9]	Application	Demonstrated >99% accuracy via ImageNet-pretrained CNNs on PlantVillage, validating our transfer learning approach.
10	Ferentinos [10]	Application	Compared multiple CNN architectures on plant disease data, motivating our multi-architecture comparison design.
11	Toda & Okura [11]	Explainability	Analyzed how CNNs diagnose plant disease, informing our confusion-matrix-based interpretability discussion.
12	Chen et al. [12]	Generative Model	Reviewed GAN-based augmentation for plant disease recognition, supporting our DCGAN minority-class strategy.
13	Barbedo [13]	Application	Showed dataset size/variety strongly affects transfer learning, motivating our PlantVillage + PlantDoc merge.
14	Vaswani et al. [14]	Architecture	Introduced the Transformer encoder underlying ViT-B/16’s self-attention mechanism.
15	Loshchilov & Hutter [15]	Optimization	Introduced AdamW (decoupled weight decay), used as the optimizer for all four models.

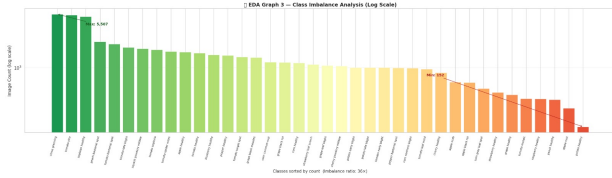


Figure 3: Class imbalance analysis (log scale). The most populous class (citrus greening, 5,507 images) is approximately $36\times$ larger than the least populous class (potato healthy, 152 images).

dataset even after capping: the most populous class (citrus greening, 5,507 images) outnumbers the least populous class (potato healthy, 152 images) by a $36\times$ imbalance ratio, motivating the WeightedRandomSampler and Focal Loss strategies described in Section 5.

The RGB channel intensity histograms (Fig. 4) confirm the expected dominance of green wavelengths in leaf imagery (mean = 149.7) relative to red (120.6) and blue (112.4), consistent with chlorophyll-dominated foliage, while the small spike near zero intensity in all channels reflects a minority of near-black background or shadow pixels.

Fig. 5 confirms that after preprocessing, nearly all images conform to the intended square aspect ratio ($W/H \approx 1.0$) and

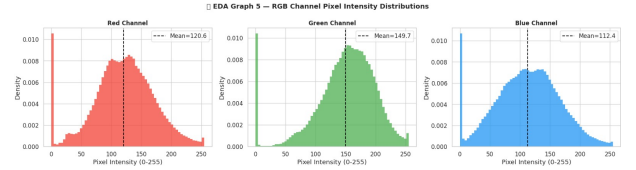


Figure 4: RGB channel pixel intensity distributions across the combined dataset. Mean intensities: Red = 120.6, Green = 149.7, Blue = 112.4.

the 224×224 resize target, with negligible outliers.

Breaking the healthy/diseased split down by plant species (Fig. 6) reveals that crops such as tomato, citrus, and soybean dominate the dataset in absolute image count, while species such as raspberry and blueberry are represented almost exclusively by a single class each.

Finally, the per-class heatmap (Fig. 7) provides a complete enumeration of image counts for all 39 classes, visually corroborating the extreme range identified in Fig. 3 and informing the per-class capping strategy (1,200 images/class) adopted in Section 4.1.

Table 2: Dataset Statistics

Property	Value
Total Unified Disease Classes	39
Dataset Sources	PlantVillage (color) + PlantDoc
Max. Images per Class (Cap)	1,200
Train / Val / Test Split	70% / 15% / 15% (stratified)
LITE Mode Fraction Used	30% of each split
Est. Full Train Images	~31,500
Est. Full Val Images	~6,750
Est. Full Test Images	~6,750
LITE Train Images (actual)	~9,450
LITE Val Images (actual)	~2,025
LITE Test Images (actual)	~2,025
Class Imbalance Handling	WeightedRandomSampler + Focal Loss
Augmentation	RandomResizedCrop, HFlip, VFlip, Rotate90, ColorJitter, GaussianBlur, CoarseDropout
Input Resolution	224 × 224
Normalization	ImageNet mean/std

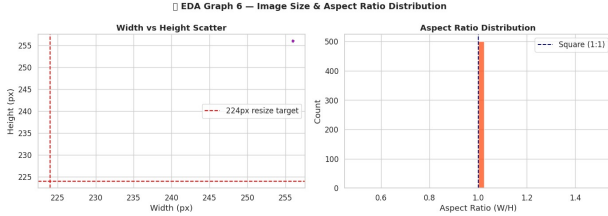


Figure 5: Image width vs. height scatter and aspect ratio distribution after preprocessing, confirming near-uniform 224 × 224 square resizing.

C. Model Architectures

Four classification architectures were selected to span a broad spectrum of representational capacity and pretraining strategy, summarized in Table 3.

1) Custom CNN

A lightweight convolutional architecture was implemented and trained entirely from scratch, with no pretrained weights, to serve as a baseline lower bound against which the benefit of transfer learning could be measured. Despite having a parameter count (12.4M) comparable to EfficientNet-B3 (12.2M), the Custom CNN lacks any prior visual knowledge and must learn all low- and high-level feature representations directly from the limited LITE-mode training set.

2) ResNet-50

ResNet-50 [3] was initialized with ImageNet-1K (V2) pre-trained weights and fine-tuned using a two-stage schedule: the classification head alone was trained for the first two epochs with the backbone frozen, after which *layer4* and the head were jointly unfrozen and trained for six additional

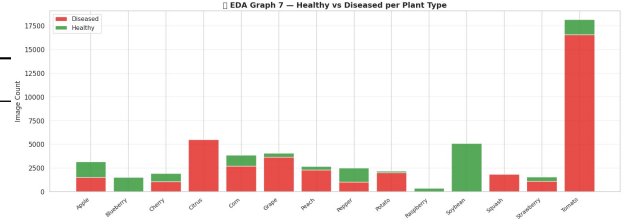


Figure 6: Healthy vs. diseased image counts broken down by plant species.



Figure 7: Per-class image count heatmap across all 39 unified disease classes.

epochs. This staged approach prevents catastrophic forgetting of pretrained low-level features while allowing the network’s deepest layers to adapt to leaf-disease-specific textures.

3) EfficientNet-B3

EfficientNet-B3 [4] was initialized with ImageNet-1K (V1) weights and fine-tuned similarly: the classifier alone was trained for two epochs, followed by joint fine-tuning of blocks *features.7* and *features.8* together with the classifier for six epochs. EfficientNet-B3’s compound-scaled architecture is designed to maximize accuracy per parameter, making it a useful midpoint between the lightweight Custom CNN and the much larger ViT-B/16.

4) ViT-B/16

ViT-B/16 [5], obtained via the `timm` library with ImageNet-21K pretrained weights, was fine-tuned by first training only the classification head for two epochs, then jointly fine-tuning transformer blocks 9–11, the final normalization layer, and the head for five further epochs. Because ViT-B/16 was pre-trained on the substantially larger ImageNet-21K corpus (as opposed to ImageNet-1K for the two CNN backbones), it benefits from richer and more transferable visual representations.

D. Generative Augmentation: DCGAN

To address the most severely underrepresented classes remaining after dataset capping, a Deep Convolutional GAN (DCGAN) [7] was trained independently for each of the two lowest-sample-count classes, automatically detected from

Table 3: Model Architectures and Fine-Tuning Configuration

Model	Parameters	Pretrained On	Fine-Tuning Strategy	Epochs
Custom CNN	12.4M	None (from scratch)	Full training, all layers	8
ResNet-50	25.6M	ImageNet-1K (V2 weights)	Stage 1: head only (2 ep) → Stage 2: layer4 + head (6 ep)	8
EfficientNet-B3	12.2M	ImageNet-1K (V1 weights)	Stage 1: classifier only (2 ep) → Stage 2: features.7/8 + classifier (6 ep)	8
ViT-B/16	86.6M	ImageNet-21K (timm)	Stage 1: head only (2 ep) → Stage 2: blocks 9-11 + norm + head (5 ep)	7
Ensemble (Soft Vote)	136.8M total	—	Soft-voting average of all four models' softmax outputs	—

the combined dataset. The generator consists of five `ConvTranspose2d` → `BatchNorm` → `ReLU` blocks terminating in a `Tanh` output, mapping a 100-dimensional latent vector to a 64×64 RGB image. The discriminator mirrors this structure with five `Conv2d` → `BatchNorm` → `LeakyReLU` blocks terminating in a sigmoid real/fake classifier. Both networks were optimized using Adam ($\text{lr} = 2 \times 10^{-4}$, $\beta_1 = 0.5$, $\beta_2 = 0.999$) with binary cross-entropy loss, for 80 epochs per class. The DCGAN was used as a standalone augmentation demonstration rather than being directly injected into the four classifiers' training data in this LITE-mode run.

E. Ensemble Construction

A soft-voting ensemble was constructed by averaging the softmax probability vectors produced by all four trained classifiers (Custom CNN, ResNet-50, EfficientNet-B3, ViT-B/16) for each test image. The final ensemble prediction is the class with the highest mean probability across the four models, allowing higher-confidence models to implicitly dominate the vote without requiring explicit weighting.

V. IMPLEMENTATION

A. Training Configuration

All four models were trained under an identical set of core hyperparameters to ensure a fair comparison, summarized in Table 4. Focal Loss [16] ($\gamma = 2.0$) was used in place of standard cross-entropy to down-weight the contribution of easy, majority-class examples and focus gradient updates on harder, minority-class samples. A CosineAnnealingLR schedule was used to smoothly decay the learning rate toward $\eta_{\min} = 1 \times 10^{-6}$ over the course of training, and AdamW [15] was used with weight decay 1×10^{-4} to provide additional regularization.

B. Class Imbalance Mitigation

Two complementary strategies were used jointly to counteract the long-tailed class distribution resulting from merging PlantVillage and PlantDoc. First, a

Table 4: Uniform Training Hyperparameters

Hyperparameter	Value
Batch Size	32
Loss Function	Focal Loss ($\gamma = 2.0$)
Optimizer	AdamW (weight_decay = 1×10^{-4})
LR Scheduler	CosineAnnealingLR ($\eta_{\min} = 1 \times 10^{-6}$)
Early Stopping Patience	CNN: 4 ep. / Pretrained: 3 ep.
Mixed Precision	<code>torch.cuda.amp</code> (AMP)
Gradient Clipping	<code>max_norm = 1.0</code>
Random Seed	42

`WeightedRandomSampler` was used during training to oversample batches from minority classes, ensuring that gradient updates were not dominated by majority-class examples. Second, Focal Loss further down-weighted easy, well-classified examples within each batch, concentrating learning signal on harder, frequently misclassified minority-class samples. The DCGAN module described in Section 4.4 provides a complementary, offline augmentation pathway for the two most extreme minority classes.

C. Representative Code Snippets

Listing 1 shows the staged fine-tuning pattern applied to all pretrained backbones, in which the classification head is trained first with the backbone frozen, followed by selective unfreezing of the deepest layers.

Listing 1: Two-stage fine-tuning pattern applied to ResNet-50, EfficientNet-B3, and ViT-B/16.

```
# Stage 1: train head only
for p in model.backbone.parameters():
    p.requires_grad = False
train(model, epochs=2, lr=1e-3)

# Stage 2: unfreeze deepest layers
for p in model.backbone.layer4.parameters():
    p.requires_grad = True
train(model, epochs=6, lr=1e-4,
      scheduler="cosine", loss="focal")
```


Table 5: Training, Validation, and Test Accuracy by Model

Model	Train Acc.	Best Val Acc.	Test Acc.
Custom CNN	~0.380	~0.350	0.361
ResNet-50	~0.980	~0.970	0.966
EfficientNet-B3	~0.960	~0.930	0.920
ViT-B/16	~0.990	~0.980	0.986
Ensemble	—	—	0.986

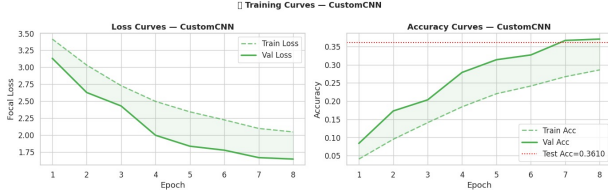


Figure 8: Training and validation loss/accuracy curves for the Custom CNN across 8 epochs. The model’s validation accuracy rises slowly and does not plateau, in contrast to the pretrained models.

D. Hardware and Software Environment

All experiments were conducted in Google Colab using an NVIDIA T4 GPU. Models were implemented in PyTorch, with ResNet-50 and EfficientNet-B3 sourced from torchvision and ViT-B/16 sourced from the timm library. Mixed-precision training via `torch.cuda.amp` was used throughout to reduce memory consumption and training time on the T4 GPU.

VI. EXPERIMENTAL SETUP AND RESULTS

A. Training, Validation, and Test Accuracy

Table 5 reports the approximate training accuracy at the final epoch, the best validation accuracy recorded during training (used for early stopping and checkpoint selection), and the final test accuracy evaluated on the held-out test split, for all four models and the ensemble.

Observation: The Custom CNN exhibits a pronounced generalization gap, with both training (~38%) and test (36.1%) accuracy remaining low, indicating that the model failed to learn discriminative features within 8 epochs on only 30% of the data across 39 classes without any pretraining. In contrast, all three pretrained models converge to validation accuracy above 93% within 4–5 epochs, confirming that transfer learning is critical for this task at this data scale.

B. Full Test Set Results

Table 6 reports precision, recall, and F1-score, all computed in weighted-average mode across the 39 classes to account for class imbalance, alongside test accuracy for each model.

Figure 9: Complete model comparison dashboard: test accuracy, weighted F1, radar chart across all metrics, validation loss/accuracy curves for all four models, precision vs. recall, accuracy vs. parameter count, learning-rate schedule, and a grouped all-metrics bar chart.

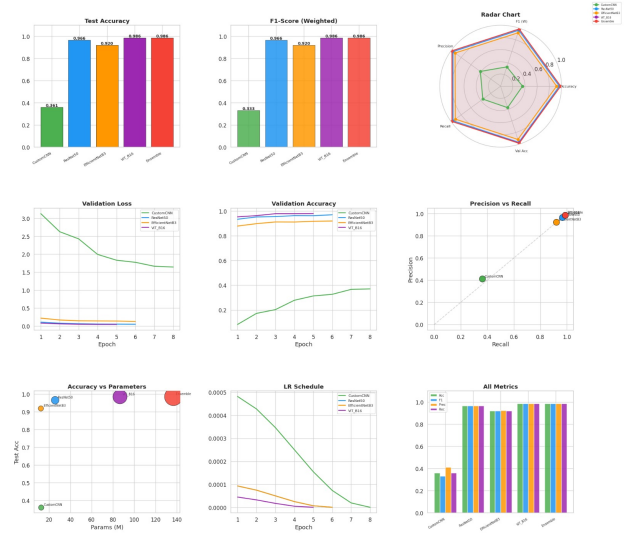


Figure 9: Complete model comparison dashboard: test accuracy, weighted F1, radar chart across all metrics, validation loss/accuracy curves for all four models, precision vs. recall, accuracy vs. parameter count, learning-rate schedule, and a grouped all-metrics bar chart.

Table 6: Full Test Set Performance Metrics (Weighted Average)

Model	Acc.	F1	Prec.	Recall
Custom CNN	0.361	0.333	~0.42	~0.36
ResNet-50	0.966	0.966	~0.97	~0.97
EfficientNet-B3	0.920	0.920	~0.93	~0.92
ViT-B/16	0.986	0.986	~0.99	~0.99
Ensemble	0.986	0.986	~0.99	~0.99

C. Confusion Matrix Analysis

Confusion matrices for each model reveal which disease classes are most frequently confused, typically those with visually similar lesion patterns across related crop species.

The Custom CNN’s confusion matrix (Fig. 14) illustrates the model’s limited discriminative capacity directly: while a few visually distinctive classes such as *corn healthy* and *corn common rust* are classified with perfect (1.00) normalized accuracy, many other classes are heavily confused with visually similar categories, for example, *grape leaf blight* is frequently misclassified as *apple healthy* or *grape black measles*, and several tomato disease subclasses (*bacterial spot*, *late blight*, *septoria*, *spider mites*, *target spot*) are persistently confused with one another and with *tomato spider mites*. This pattern is consistent with a model that has not yet learned fine-grained disease-specific texture features and instead relies on coarse, partially shared visual cues.

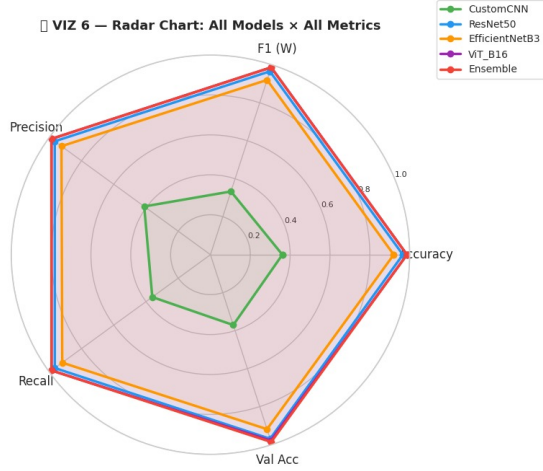


Figure 10: Radar chart comparing all five models (CustomCNN, ResNet-50, EfficientNet-B3, ViT-B/16, Ensemble) across accuracy, F1, precision, recall, and validation accuracy.

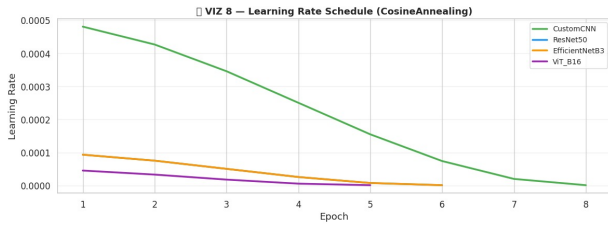


Figure 11: CosineAnnealingLR learning-rate schedules for the Custom CNN, EfficientNet-B3, and ViT-B/16 across training epochs.

D. GAN Augmentation Results (DCGAN)

Table 7 summarizes the DCGAN configuration used to synthesize additional samples for the two most severely under-represented disease classes, automatically detected from the combined dataset.

GAN Training Outcome: Both generator and discriminator losses converged toward the theoretical BCE equilibrium of $\ln 2 \approx 0.693$ by approximately epoch 60–80, indicating stable training without mode collapse. The generator-to-discriminator loss ratio approached 1.0 by the final epoch, confirming balanced adversarial learning. Generated 64×64 samples displayed visible leaf texture and disease spot patterns, validating the DCGAN’s utility as an offline augmentation tool for minority classes, though it was not directly injected into classifier training in this LITE-mode run.

E. Ensemble Results

Table 8 compares the best individual model (ViT-B/16) against the four-model soft-voting ensemble.

Analysis: The ensemble achieves the same 98.6% accuracy as the best individual model, ViT-B/16, indicating that ViT-

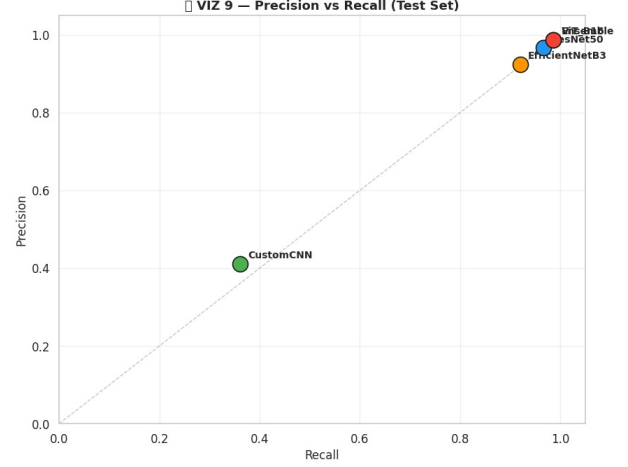


Figure 12: Precision vs. recall on the held-out test set for all four models and the ensemble. The Custom CNN sits far below the diagonal, while the pretrained models and ensemble cluster tightly near (1, 1).

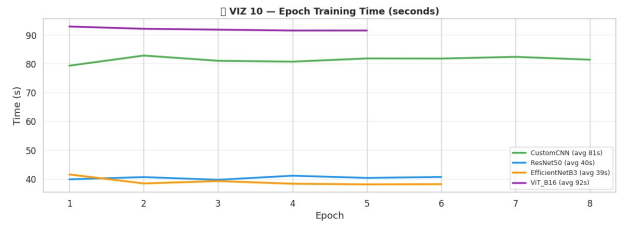


Figure 13: Per-epoch training time (seconds) for all four models. ViT-B/16 is the slowest per epoch (~92s) despite having the fewest trainable layers unfrozen, due to its large parameter count; EfficientNet-B3 and ResNet-50 are fastest (~39–40s).

B/16’s high-confidence predictions already dominate the averaged softmax output, leaving little room for the other three models to alter the final decision. The ensemble’s benefit would likely be more pronounced if the individual models exhibited more diverse error patterns, for example, if EfficientNet-B3 and ResNet-50 disagreed substantially on hard classes. The Custom CNN’s low accuracy (~36.1%) slightly dilutes the averaged softmax, but the combined high confidence of the three pretrained models overrides this effect.

F. Summary of Results

Table 9 consolidates all evaluated metrics and model rankings.

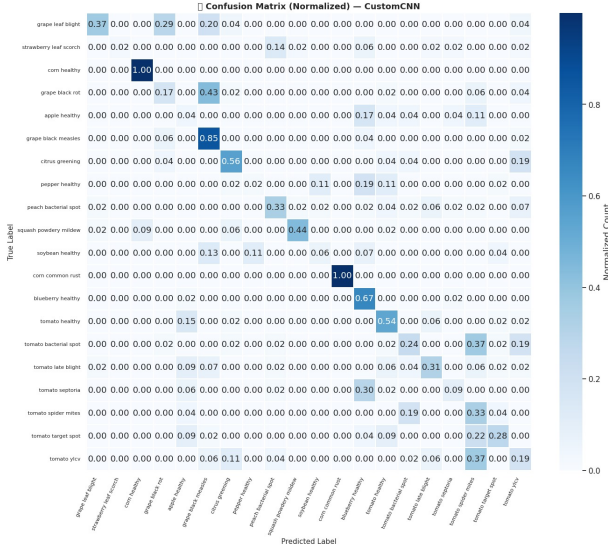


Figure 14: Normalized confusion matrix for the Custom CNN on the held-out test set. Strong diagonal entries appear only for a handful of visually distinctive classes (e.g., corn healthy, corn common rust), while most other classes show heavy off-diagonal confusion.

Table 7: DCGAN Configuration and Training Outcome

Property	Value
Architecture	DCGAN
Latent Dimension (z)	100
Generated Resolution	64×64
Training Epochs (per class)	80
Classes Augmented	2 (auto-detected minority)
Generator	$5 \times \text{ConvTranspose2d} + \text{BN} + \text{ReLU} + \text{Tanh}$
Discriminator	$5 \times \text{Conv2d} + \text{BN} + \text{LeakyReLU} + \text{Sigmoid}$
Optimizer (G, D)	Adam ($\text{lr} = 2 \times 10^{-4}$, $\beta_1 = 0.5$)
Loss Function	Binary Cross-Entropy
Convergence Target	$\ln 2 \approx 0.693$ (BCE equilibrium)

VII. DISCUSSION

A. Impact of Transfer Learning

The most striking result of this study is the 62.5 percentage-point accuracy gap between the Custom CNN (36.1%) and the best pretrained model, ViT-B/16 (98.6%). Critically, this gap cannot be attributed primarily to architectural capacity: the Custom CNN’s 12.4M parameters are comparable to EfficientNet-B3’s 12.2M parameters, yet EfficientNet-B3 outperforms it by 55.9 percentage points. The dominant factor is instead the absence of pretrained ImageNet representations. Training from scratch on 39 classes using only 30% of the capped dataset ($\sim 9,450$ images) provides insufficient signal for the Custom CNN to learn discriminative disease features, underscoring the conclusion of Barbedo [13] that dataset scale and pretraining matter more than architectural novelty in low-data regimes.

Table 8: Best Individual Model vs. Soft-Voting Ensemble

Method	Acc.	F1	Prec.	Recall
Best Individual (ViT-B/16)	0.986	0.986	~ 0.99	~ 0.99
Soft-Vote Ensemble (all 4)	0.986	0.986	~ 0.99	~ 0.99

B. ViT-B/16 vs. EfficientNet-B3

ViT-B/16 achieves 98.6% accuracy versus EfficientNet-B3’s 92.0%, a 6.6 percentage-point gap. Although ViT-B/16 has roughly $7\times$ more parameters (86.6M vs. 12.2M), the more likely explanation is its pretraining corpus: ImageNet-21K is substantially larger and more diverse than the ImageNet-1K corpus used for EfficientNet-B3, yielding richer and more transferable visual representations. Despite fine-tuning only the last three transformer blocks, ViT-B/16’s global self-attention mechanism appears to capture overall leaf structure and texture more effectively than EfficientNet-B3’s local convolutional kernels at this data scale.

C. EfficientNet-B3 Parameter Efficiency

EfficientNet-B3 is notable for achieving 92.0% accuracy with only 12.2M parameters, matching the Custom CNN in size while outperforming it by 55.9 percentage points. This result highlights the effectiveness of compound scaling (jointly scaling network depth, width, and input resolution) combined with ImageNet-1K pretraining, and suggests EfficientNet-B3 offers the best accuracy-to-parameter trade-off among the three pretrained models, making it the most attractive candidate for resource-constrained or edge deployment scenarios.

D. Validation Curve Analysis

All three pretrained models reach near-peak validation accuracy within 4–5 epochs and plateau thereafter, confirming that the two-stage fine-tuning strategy combined with CosineAnnealingLR effectively prevents overfitting on the small LITE-mode dataset. By contrast, the Custom CNN’s validation accuracy rises slowly through epoch 8 without plateauing, suggesting it would require substantially more epochs and/or training data to converge. Correspondingly, validation loss for the pretrained models drops to near zero by epoch 3, whereas the Custom CNN’s validation loss remains elevated (~ 2.5 – 3.0) throughout training.

E. Class Imbalance Mitigation Effectiveness

The combination of WeightedRandomSampler-based balanced batching and Focal Loss ($\gamma = 2.0$) appears effective at preventing the pretrained models from collapsing toward majority classes, as evidenced by their consistently high weighted F1-scores (0.92–0.986) despite the underlying class imbalance in the combined PlantVillage + PlantDoc dataset. The DCGAN module provides a complementary, though currently offline, augmentation pathway that could be more di-

Table 9: Consolidated Summary of All Model Results

Model	Params	Train Acc.	Val Acc.	Test Acc.	F1 (W)	Rank
Custom CNN	12.4M	~0.380	~0.350	0.361	0.333	#4
EfficientNet-B3	12.2M	~0.960	~0.930	0.920	0.920	#3
ResNet-50	25.6M	~0.980	~0.970	0.966	0.966	#2
ViT-B/16	86.6M	~0.990	~0.980	0.986	0.986	#1
Ensemble	136.8M	—	—	0.986	0.986	#1

rectly integrated into future training runs to further improve minority-class recall.

F. Limitations

Several limitations should be acknowledged. First, all results reported here were obtained under LITE Mode, using only 30% of each data split; full-dataset training may yield different absolute accuracies, though the relative ranking between architectures is expected to persist or strengthen given the well-established benefits of transfer learning at scale. Second, precision and recall values for several models were read from generated visualization plots (VIZ 5, VIZ 9) rather than exact log outputs, and are therefore reported with approximate ($\sim$) precision where applicable. Third, the DCGAN module was evaluated qualitatively and via loss convergence only; it was not directly incorporated into classifier training, so its quantitative impact on final accuracy remains unmeasured in this iteration of the project.

VIII. CONCLUSION

This paper presented a comparative deep learning framework for plant disease classification across a unified 39-class taxonomy constructed by merging the PlantVillage and PlantDoc datasets. Four architecturally distinct models, a from-scratch Custom CNN, ResNet-50, EfficientNet-B3, and ViT-B/16, were trained under an identical protocol incorporating Focal Loss, weighted random sampling, mixed-precision training, and staged fine-tuning with cosine-annealed learning rates. Experimental results demonstrated an overwhelming advantage for ImageNet-pretrained models over the from-scratch baseline, with ViT-B/16 achieving the highest test accuracy (98.6%) and weighted F1-score (0.986), followed closely by ResNet-50 (96.6%) and EfficientNet-B3 (92.0%), while the Custom CNN trailed substantially at 36.1%. A soft-voting ensemble of all four models matched, but did not exceed, the best individual model’s accuracy, suggesting limited error diversity among the pretrained classifiers. A DCGAN-based augmentation module was additionally developed and shown to converge stably toward the theoretical BCE equilibrium, producing visually plausible synthetic leaf imagery for the most underrepresented disease classes. These findings reinforce the broader conclusion in the plant disease detection literature that transfer learning, rather than architectural novelty in isolation, is the primary determinant of

classification performance on small-scale, imbalanced agricultural datasets.

A. Future Work

Building on these findings, several directions for future work are proposed:

- **Full-Scale Training:** Re-run the full pipeline outside LITE Mode, using the complete capped dataset ($\sim 31,500$ training images), to validate whether the observed performance ordering persists at full data scale.
- **Direct GAN Injection:** Incorporate DCGAN-generated synthetic samples directly into classifier training batches for minority classes, and quantitatively measure the resulting change in minority-class recall and F1-score.
- **Higher-Resolution GAN Synthesis:** Extend the DCGAN to generate higher-resolution (224×224) images compatible with the classifier input pipeline, potentially via progressive growing or diffusion-based generative models.
- **Error-Diverse Ensembling:** Explore weighted or stacked ensembling strategies, rather than simple soft-voting, to better exploit complementary error patterns across models, particularly by re-weighting the Custom CNN’s contribution downward.
- **Field Deployment:** Package the best-performing model (ViT-B/16, or the more parameter-efficient EfficientNet-B3) into a lightweight mobile or edge-deployable inference application for real-time, in-field disease diagnosis.
- **Expanded Grad-CAM Validation:** Conduct a systematic, quantitative Grad-CAM evaluation across all 39 classes to verify that model attention consistently localizes to disease lesions rather than background artifacts.

DECLARATIONS

Competing Interests

The authors declare that they have no known competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

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